Thuvaragan Sooriyakumaran

PhD Student · Designer · Developer

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From planet earth, loves to integrate art, science and technology to create a smile in the face of others. Moderate experience in systems programming, robotics, digital design, algorithms, and performance optimization.

Complimentary experience in web development, cross-platform app development, audio manipulation, and graphics manipulation. Currently obsessed with modular robotics, RISC-V IP design, open-source toolchains, and devops of corresponding workflows.

Experience

Research Assistant - Internship — Remote - Singapore

College of Computing & Data Science (CCDS), NTU — 2024 Dec - 2025 Jul

- C99 Bundle adjustment implementation
- QEMU, LiteX-Sim, Chipyard, and ZCU-102 compilation and workflows
- Literary review of vector processing options for Rocket core
- Cross Core shared memory workflows

Education

PhD Student — Singapore

College of Computing & Data Science (CCDS), Nanyang Technological University (NTU) — Aug 2026 - Present

BSc. in Engineering - Electronics and Telecommunication — Colombo, SriLanka

University of Moratuwa | OGPA - 3.75 | First Class — 2021 - 2026

Charted Global Management Accounting (CGMA) — Colombo, SriLanka

Achievers Lanka | Certificate Level Completed — 2021

System {Verilog} for ASIC/FPGA Design & Simulation — Colombo, SriLanka

Short course by ENTC department UoM with Skillsurf.lk — Jan 2023 - May 2023

- Assignment 3 - ASIC flow report for MVM UART using SAED 32nm EDK & Synopsys design compiler
- Assignment 4 - FPGA implementation and demo of MVM UART

Secondary Education — Jaffna, SriLanka

Jaffna Hindu College | G.C.E. OL - 9A | G.C.E. AL - 3A — 2011 - 2020

Skills

Programming C/C++, **System{Verilog}**, **Python**, **TypeScript**, **Zig**, Rust, Lua, Nix

Languages **English**, **Tamil**, Japanese, Sinhala, French

Tools **Vivado / Verilator**, **ROS**, KiCAD / Altium Designer, SolidWorks, MATLAB

Projects

Qwen3-0.6B inference pipeline on custom RV32IM core in FPGA — thuvasooriya/ip_gemma

DVCON 25 - India: 1st Runner's Up — May 2025 - Sep 2025

- Research and Implementation project for DVCON25 to design a novel resource efficient accelerator IP for a custom RV32IM core
- Ran whole inference pipeline of Qwen3-0.6B with Q8 quantization and custom token embeddings

ViT Accelerator IP for a custom RISC-V core in FPGA — thuvasooriya/vit-malware-detector

DVCON 24 - India — May 2024 - Sep 2024

- Research and Implementation project for DVCON24 to design a novel ViT accelerator IP for a custom RISC-V core
- Selected in the 20 teams from >100 teams in Stage 1

Image segmentation research and evaluation for bin-picking — mora-bprs/sam-model

Semester 4 group project — March 2024 - June 2024

- Extensive literary review and recreation of image segmentation and CNN model papers in python
- Development of simple gripper mechanism and associated PCB for testing
- Extensive testing, benchmarking and documentation

Smart plug product research and development — thuvasooriya/pluggu

Semester 2 group project, Short course group project — Mar 2023 - May 2023

- Smart plug with remote control through LAN and WAN, developed around ESP32-S3 micro controller
- Another smart plug design and implementation was followed as part of a short course

STM32 two-wheeled line follower robot with simple manipulator

team-itro/sem3slrc

Semester 3 group project — Aug 2023 - Dec 2023

- Prototyped, designed and implemented a robot that is capable of following white, colored lines
- Custom arm mechanism was designed and integrated to achieve required object manipulation

Portable doctor companion device with SBC

biosense-ai/biosense-ai-web-server

Mecha 23 - 2nd Runner Up — Aug 2023 - Oct 2023

- Integrating ECG monitor and SpO_2 sensor with machine learning to make screening process easier for doctors
- Orange-Pi Zero 2W SBC is used with custom Linux to integrate monitoring, web-server and prediction algorithms

Maze Solving Micromouse

team-itro/kitro

RoboFest 24 — Sep 2024 - Dec 2024

- Maze Solving micromouse using STM32F411 microcontroller.
- Participated in RoboFest 24, Research on 2nd prototype ongoing.

Analog ECG monitor PCB development

Semester 3 group project — Aug 2023 - Oct 2023

Semester 5 Group Projects

Demonstrated as per corresponding modules — Sep 2024 - Dec 2024

- RISC-V RV32I pipeline-processor research and implementation
- Single Precision FPU IP research and implementation
- Robot manipulator with 4 DOF

Semester 7 Group Projects

In Progress as per module requirements — Sep 2025 - Present

- ROS2 Labs and cross-platform testing
- BitSerial Bus implementation with 3 masters and 3 slaves
- MNIST Co-Processor-Accelerator IP implementation
- Promoter Analysis

Achievements

Lead **Deputy Batch Representative**, Engineering Faculty, University of Moratuwa,
Senior(Head) Prefect, 2020, Jaffna Hindu College

Awards **Right To Information Act, National Debate - 1st Place**, Tamil, Team Lead,
All Island Senior Dialog Drama - 3rd Place

References

Dr. Ranga Rodrigo ranga@uom.lk

B.Sc. Eng. Hons. (Moratuwa), M.E.Sc. (Western, Canada), Ph.D. (Western, Canada), SMIEEE

Senior Lecturer, University of Moratuwa

Dr. Thayaparan Subramaniam thayaparan@uom.lk

B.Sc.Eng (Peradeniya), Ph.D (HKU)

Senior Lecturer, University of Moratuwa

Dr. Chamira U. S. Edussooriya charmira@uom.lk

B.Sc.Eng. (Moratuwa) M.A.Sc., Ph.D. (UVic), MIEEE

Senior Lecturer, University of Moratuwa

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Principal, Jaffna Hindu College | LLB Law Hons.